

Connor Hartley

Full Stack Software Engineer

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EXPERIENCE

Moonsworth, Remote — Full Stack Software Engineer

FEB 2023 - PRESENT

I work in the client development team responsible for designing, building and bug fixing the new developer API that controls how the Moonsworth client interacts with game servers.

My key responsibilities include the following:

- Built an SDK using Protobufs in Java to allow third-party developers to easily drive OpenGL rendered cosmetics and other content used by 500K+ MAUs.
- Implemented a solution for dynamic dependency management which allowed the software to be deployed on any platform.
- Preparing the project for open sourcing and using advanced multi-project Gradle features for Kotlin DSL build scripts, such as version catalogs, custom configurations, code styling and jar-in-jar bundling.
- Writing javadocs and external documentation in markdown, allowing hundreds of interested parties to contribute effectively.
- Creating technical specifications and organizing these plans into Agile sprints on YouTrack. Also used YouTrack to track internal bugs.
- Reviewing internal and external pull requests & providing feedback to maintain code standards/reduce technical debt.

Hideaway Games, Remote — Full Stack Software Engineer

FEB 2023 - PRESENT

I work in the plugin and web development teams responsible for designing, building and bug fixing gameplay features and the public website for Hideaway.

My key responsibilities include the following:

- I created the player housing system for Hideaway in Java & Go, which is designed to load instances of player worlds across different physical machines that players can move between (connected by a custom TCP reverse proxy). The instances are uploaded and downloaded to and from S3.
- Implemented high concurrency Spring microservices to store

SKILLS

General

- Teamwork & Leadership
- Problem-solving
- Agile
- Java [8 years]
- JavaScript [7 years] (NodeJS, Typescript)
- GoLang [2 years]
- Kubernetes
- Docker
- SQL (MySQL, PostgreSQL)
- Git (GitHub/GitLab)
- CI/CD (GitHub Actions, Jenkins, CircleCI)
- AWS (EBS, Amplify, RDS)
- Cloudflare (DNS, Pages, Workers)
- RabbitMQ, NATS
- Protobuf
- YouTrack
- Writing User & Dev Documentation

Java [8 years]

- RxJava
- Spring
- Guice, Guava, Gson
- Redis (Jedis)
- JDBI, HikariCP
- Gradle (Groovy, Kotlin)
- Maven
- JUnit
- SpongePowered Mixins
- OpenGL (GLSL)

Web [7 years]

- NodeJS
- JavaScript, TypeScript
- HTML, CSS (Sass)

persistent data for various game features (such as player items, player statistics & player achievements) on SQL and Redis. This allows us to handle 1000's of concurrent players at once over a network deployed on Kubernetes.

- Using SpongePowered Mixins and Git Patches to improve the performance of the game software or add additional functionality for gameplay features that are made using Guice, Guava and GSON.
- Forked and reworked a Go program that converts the game's native world format into a custom binary world format to work with the latest versions of the game.
- Made several improvements to the public facing website using TypeScript, React and Sass.

- VueJS, ReactJS, HyperX, Svelte
- AstroJS
- Prisma
- GhostCMS
- Puppeteer

LANGUAGES

English

Voldex Games, Remote — Web/Tooling Developer

JUL 2022 - FEB 2023

Voldex is one of the largest UGC gaming companies having raised over \$40M from top VC's. I worked in the web/tooling development team responsible for designing, building and bug fixing various web applications and the public website for Voldex Games.

My key responsibilities include the following:

- I created a new quality assurance panel for Voldex Games. I used NodeJS for tracking issues, features, the testing session duration and conversations between the developers. This increased the QA team's productivity by 10x.
- I created the public Voldex Games website from scratch (<https://voldex.com>) & deployed on Cloudflare Workers using AstroJS, VueJS, Sass and TypeScript. It generated approximately 10k impressions monthly.
- Using Typescript and Puppeteer I created data scraping tools allowing Voldex to track other games' growth on the Roblox platform to help them make better acquisition decisions. The data was stored into Airtable to analyze.
- Used cloud technologies such as AWS Amplify, AWS ElasticBeanstalk, Cloudflare Workers and CloudFlare Pages to deploy and scale the dashboards and tools I made.

Voldex Games (Mineteria), Remote — Full Stack Software Engineer

JAN 2017 - JUL 2022

I worked in the plugin development team responsible for designing, building and bug fixing gameplay features for Mineteria.

My key responsibilities include the following:

- Ensuring the network can handle thousands of concurrent players. Built the core microservice architecture with Kubernetes on bare metal & wrote the core services that ensure data

consistency across hundreds of game servers (leveraging RabbitMQ & Redis).

- I refactored an in-house framework for quickly developing minigames and launched two new games built on top of that framework that both reached 100K MAU at peak.
- Created various integrations for cross-platform support as one of the first ever cross-platform Minecraft servers. This included changing the orientation of certain game objects depending on the protocol, changing the behavior of the game servers anti-cheat and fixing player movement synchronization.
- I worked on various performance improvements to inhouse and external libraries that included implementing various layers of data caching to improve protocol remapping speeds, improving the behavior of the proxy to handle game server switching and performance improvements to multi-version protocol remapping using custom data structures & object pooling.
- Managed a team of engineers by reviewing their work, meeting up frequently to ensure they were meeting their targets and provided assistance when needed.

EDUCATION

UCOL, Palmerston North — *Diploma in Information Technical Support*

FEB 2016 - DEC 2018

Freyberg High School, Palmerston North — *NCEA Level 3*

FEB 2014 - DEC 2017